Apache Cerebrum: Flexible tool for Constructing Computational Neuroscience Models from Large Public Databases and Brain Atlases.

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Introduction

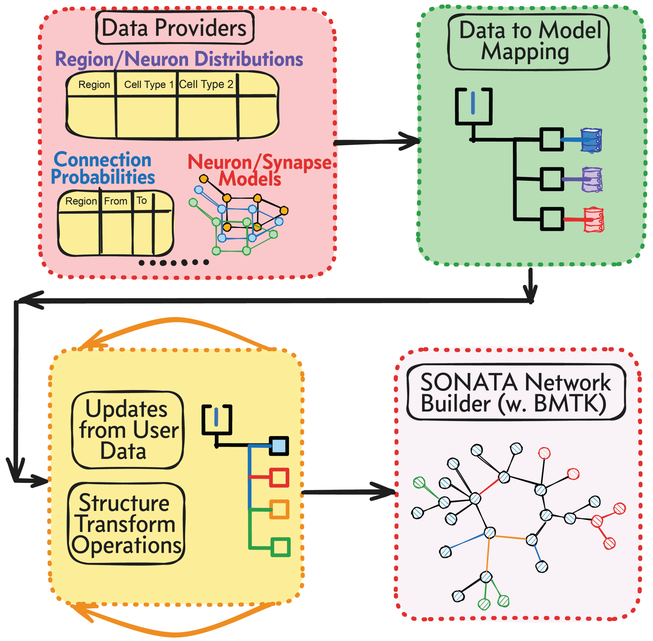
Large compendiums of datasets compiled in public databases such as whole-brain atlases provide extensive and detailed maps of the brain. However, using such brain atlases for the construction of computational models remains a challenge. Diverse databases are required for construction of detailed models and becomes difficult to modify and reuse.

Spatial transcriptomics data allows for regional and layer-wise distributions of the different cell types. Synaptic physiology databases provide connection probabilities between the sets of neuron classes. Data from all these datasets need to incorporated with along with the model databases to construct detailed models such as V1 model from Allen institute [1]. In this work, we developed a flexible and extendable tool in python that allows queries over different databases and brain atlases and applies transformations the resultant data. The results of such processing can be directly incorporated within the computational models described using SONATA format.

Overall Workflow in Cerebrum

Cerebrum workflow proceeds as follows:

→ Extract data from various experimental and brain atlas datasets using "Data Provider" classes that query the relevant databases.
→ Map acquired data to regions of interesting using the "Data2Model" module.
→ Fill-in any missing details of the model via an optional user-defined custom modifications.
→ Realize the final network of neurons in SONATA format from the model description.



References

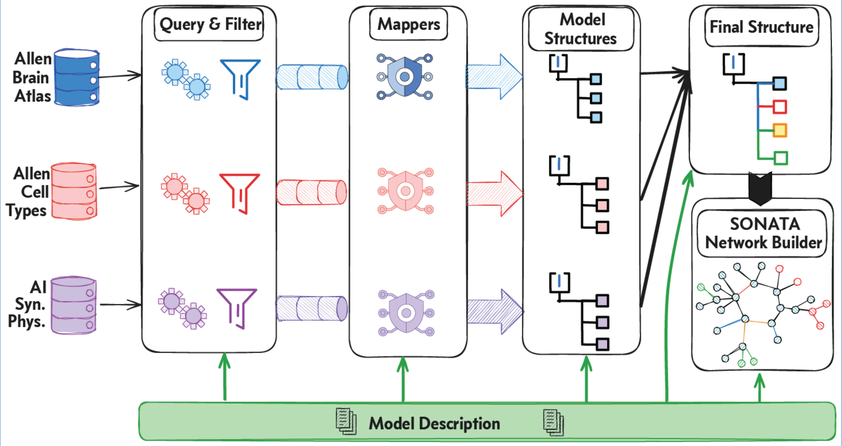
1. Billeh, Y. et al. Systematic Integration of Structural and Functional Data into Multi-scale Models of Mouse Primary Visual Cortex Neuron106.3, 388–403.e18 (2020).
2. Zhang, M. et al. Molecularly defined and spatially resolved cell atlas of the whole mouse brain. Nature 624, 343–354 (2023).
3. Marru, S. et. al. Cybershuttle: An End-to-End Cyberinfrastructure Continuum to Accelerate Discovery in Science and Engineering. ACM PEARC '23, 26–34 (2023).

Model Description

Cerebrum uses configuration files (a set of Json files) for various steps in construction of model, which includes the following sections:

❖ **Data Providers**: Links to source database and query parameters
❖ **Filter and Transformers**: Filter and selection criteria on the queried data.
❖ **Data2Model Mappers**: Mapper methods that map data to model components.
❖ **User Modifications**: User Inputs that modify the data-derived model.

The output is a computational model in the SONATA format.



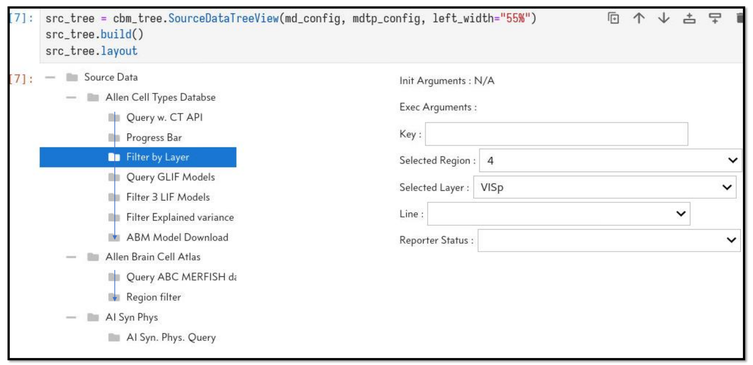
All the steps can be modified through a plugin in Jupyter or editing JSON files.

Source Data Providers

Source Data provider module includes:

- Methods for defining the querying from the databases and filter the data based on specific criteria.
- Utilities to visualize the data provider configurations in easy-to-understand explorer view inside Jupyter notebook with the relevant parameters displayed in the side panel.

An example for construction of Layer 4 of Mouse V1 is shown below with three different data providers: Allen Cell Types database, Allen Brain Cell Atlas and AI Synaptic Physiology Database.



Each leaf in the tree is a step in the corresponding data processing workflow

Acknowledgements

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Data2Model Operations

Data downloaded from the databases are store in Json format which can be loaded as pandas data frames. Mouse V1 Layer 4 connectivity probabilities from the AI Synaptic Physiology Database is show on the right.

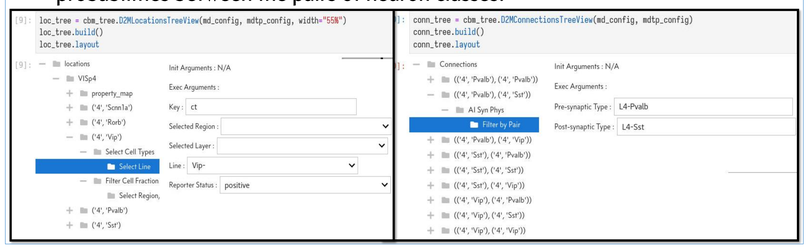
	L4-Pvalb	L4-Sst	L4-Pyr	L4-Vip
L4-Pvalb	0.610531	0.065354	0.356604	0.101648
L4-Sst	0.392124	0.016196	0.287637	0.306504
L4-Pyr	0.198113	0.045855	0.125985	0.049624
L4-Vip	0.025412	0.188560	0.000000	0.034990

Data2Model module provides:

- Customizable python functions to map atlas and other datasets to be used in different components of the models.
- Ability to integrate different data sources to map to a component in the network model.
- Two different mappings: Region Mapper and Connection Mapper.

An example mappings for Layer 4 of Mouse V1 is shown below.

❑ **Locations/Regions:** Cell types database and the MERFISH atlas data are mapped to neuron models and the distribution of neuron types, respectively.
❑ **Connections:** AI synaptic physiology data is mapped to the connection probabilities between the pairs of neuron classes.



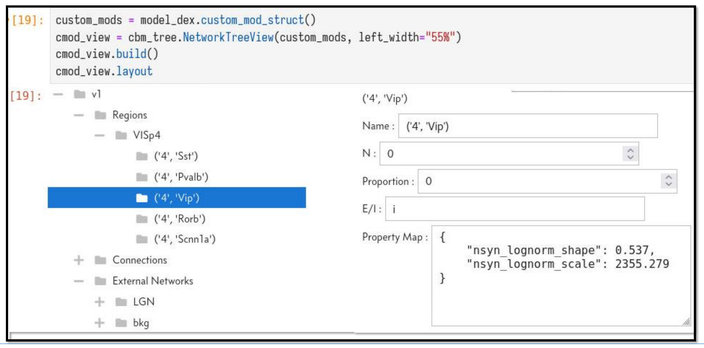
Custom User Modifications

Custom modifications module:

- Users can provide details required for network construction that are either not available in the linked databases or over-ride data-derived model.
- Utilities to visualize these modifications in Jupyter Notebook.

For the Layer 4 of Mouse V1, custom modifications include

❑ Dimensions of each region in the model.
❑ Connection Parameters not available with the AI Syn. Phys. database.
❑ Details of the networks that are external to V1 – LGN and the background networks.



Availability

Source code: https://github.com/apache/airavata-cerebrum
V1L4 Accessible via cybershuttle.org

- Offers extensibility and flexibility for diverse applications.
- Supports executions on various computational resources.